

BloodHound RCEV

Remotely Controlled Electric Vehicle

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What is the Bloodhound?

- A repurposed Kawasaki 450 ATV frame converted into a Remotely Controlled Electric Vehicle (RCEV)
- Designed to operate in hazardous environments without putting a human operator at risk
- Powered by a 52V LiFePO4 battery with dual 48V brushless hub motors (front-wheel drive)
- Steering is handled by a DOCYKE power steering servo mounted directly to the steering column



Hardware Overview

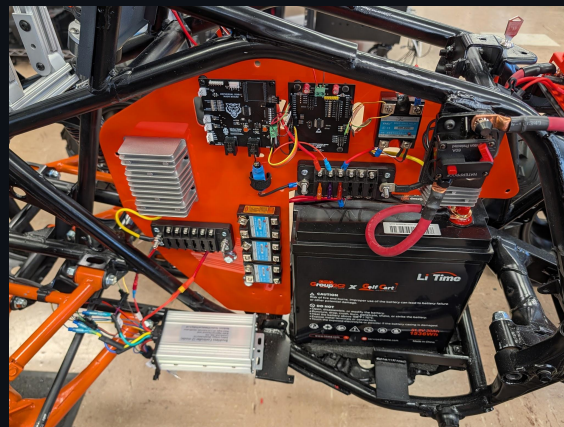
Vehicle Platform

- Kawasaki 450 frame, suspension & wheels
- 2× 48V brushless hub motors
- 2× 2-mode brushless speed controllers
- DOCYKE S350 power steering servo



Electronics

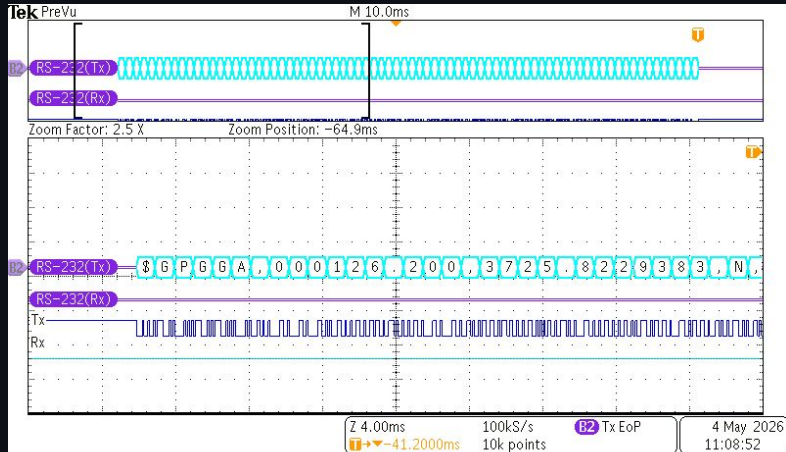
- 52V LiFePO4 battery (30Ah)
- DC/DC converter: 52V → 24V for servo & 5V step
- DC/DC converter: 52V → 12V for headlights
- 3 custom PCBs: URC, Master Main Board, Slave Dev Board
- All PIC16F1788 / PIC16LF1789 microcontrollers



RTK Hardware

Current State of RTK

- Base Station Set-up
- Communication Between Each PIKSI
- Data Packets to Slave



Three Board Architecture

URC

Universal Remote Controller
PIC16LF1789 (44-pin)

UART 115,200 bps
Wireless

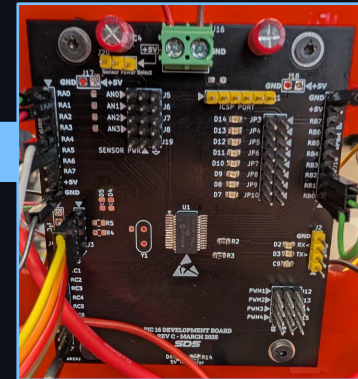
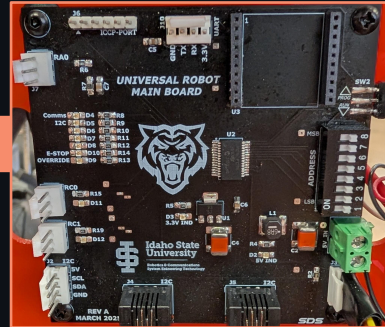
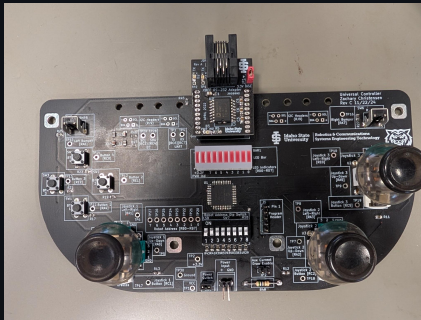
MASTER

Master Main Board
PIC16F1788 (28-pin)

I2C 100kHz
Wired

SLAVE

Slave Dev Board
PIC16F1788 (28-pin)



Outputs from Slave Board:

PWM → Motor Controllers

PWM → DOCYKE Servo (steering)

Brake & Direction signals

Data Flow: URC → Master → Slave

- **Step 1 — Operator moves joystick on the URC**

Joystick position captured as 0–255 analog value; packaged into a 6 byte UART packet

- **Step 2 — Packet travels over wireless UART at 115,200 bps to the Master board**

Packet includes: sync byte (0x24), robot address, command type, and 3 data bytes

- **Step 3 — Master decodes the packet and every 96ms sends I2C data to the Slave**

I2C transaction: START → address (0x20) → 4 data bytes → ACK each → STOP

- **Step 4 — Slave receives I2C bytes, interprets joystick values, and updates outputs**

After all 4 bytes arrive, I2C_RX_COMPLETE flag triggers motor and servo updates

Data Packet Layout					
Sync	Address	Command	Byte 1	Byte 2	Byte 3
\$(0x24)	Robot Address	J	Joystick 1 Up-Down High Byte (Left)	Joystick 1 Right-Left High Byte (Left)	Joystick 2 Up-Down High Byte (Right Lower)
\$(0x24)	Robot Address	j	Joystick 2 Right-Left High Byte (Right Lower)	Joystick 3 Up-Down High Byte (Right Upper)	Joystick 3 Right-Left High Byte (Right Upper)
\$(0x24)	Robot Address	B	Button Data Byte 1 (See Button Data Bit Map)	Button Data Byte 2 (See Button Data Bit Map)	0x00 (Not Used)

I2C Data Packets Explained

Byte 1

JOY1_UD

Drive Throttle
(Joystick 1 Up/Down)

Byte 2

JOY1_LR

Unused
(future use)

Byte 3

JOY2_UD

Brake trigger
(Joystick 2 Up/Down)

Byte 4

JOY2_LR

Steering
(Joystick 2 Left/Right)

How the values are interpreted:

- Each joystick axis arrives as a byte: 0 = full left/back, 128 = center, 255 = full right/forward
- Slave compares values to thresholds (Upper: 160, Lower: 80) to determine direction
- Servo PWM (full right= 1.2ms, center=1.5ms, full left=1.9ms)
- Values outside the deadband are mapped proportionally to motor duty cycle or servo position

RTK Software

Software Requirements:

- Swift Navigation Piksi Console
- Piksi-Piksi Communication: SBP @ 57600
- Piksi-Slave Communication: NMEA @ 9600

COM7(PK25388) Piksi Console v0.30.9

Tracking Solution Baseline Observations Settings Firmware Update Advanced

Reload Save to Flash Reset to Defaults

Show Advanced Settings: ☐

Setting

Name: uart_uartb.mode

Value: NMEA

Description: Configure mode for UART

Default value: SBP

Notes:

Console Log CSV log JSON log C:\Users\Micas\SwiftNav Log level filter: WARNING

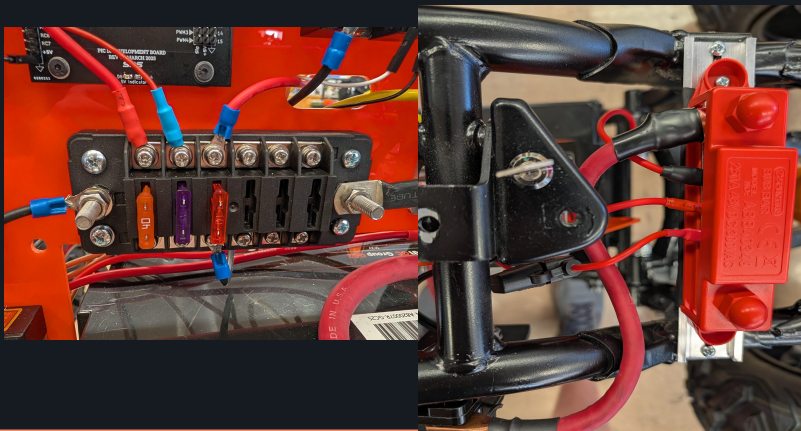
Host timestamp	Log level	Message
May 04 2026 1...	CON...	Number of tracking channels changed to 11
May 04 2026 1...	CON...	Number of settings loaded 77
May 04 2026 1...	CON...	Loaded settings vaml file from path settings.vaml

PORT: COM7 FIX TYPE: None #SATS: 0

Safety

BloodHound Hardware

- In-line fuses for 24V & 12V hubs
- In-line key switch disengaging speed controllers
- In-line Battery Pos Terminal kill switch



I2C Watchdog (Slave)

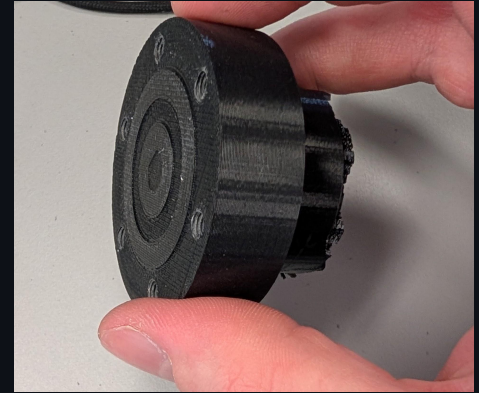
- Timer 1 counts time between I2C data packets
- No packet in ~260ms → I2C_COM_TIMEOUT
- Immediately engages brakes and stops all motors
- Prevents runaway if Master board fails

Achievements



Complications

- Broke multiple DOCYKE servos during testing
- Broke DOCYKE servo plastic coupler (twice)
- RTK Communication between Piksi and Slave Board



Cost Analysis

Team Member	Hours/ day	Total Days	Total Hours	Cost/ hr	Total
Noah Holloway	~5.5	31	170.5	\$50	\$8,525
Micah Spencer	~4	10	40.5	\$50	\$2,025
Total	9.5	41	211		\$10,550
Total + BOM					\$13,671.50

Future Expansion

Planned Additions

- Backup camera & backup lights (automatically on reverse)
- Accelerometer for orientation data
- Chemical sensors (CO and CO₂)

Thank you — Questions?